RideVault — Relational Database Schema (14-Table BCNF PostgreSQL · Crow's-Foot Notation)

Boyce-Codd Normal Form · Shared-PK Inheritance (users → riders / drivers) · Immutable Audit Logs · Double-Entry Wallet Ledger

Primary Key (PK) · Foreign Key (FK) · NOT NULL · Nullable · ||——< 1 : N · ||—|| 1 : 1

refresh_tokens

token_id	BIGSERIAL	PK
user_id	BIGINT	FK
token_hash	VARCHAR(500)	
expiry_date	TIMESTAMP	
revoked	BOOLEAN	
created_at	TIMESTAMP	

▶ Indexes: PK, UQ(token_hash), idx_user

ride_status_logs

log_id	BIGSERIAL	PK
ride_id	BIGINT	FK
from_status	VARCHAR(20)	
to_status	VARCHAR(20)	
changed_by_user_id	BIGINT	FK
changed_at	TIMESTAMP	
reason	VARCHAR(255)	

▶ Indexes: PK, idx_ride_status_logs_ride

ride_ratings

rating_id	BIGSERIAL	PK
ride_id	BIGINT	FK
stars	INTEGER	
comment	VARCHAR(255)	
created_at	TIMESTAMP	

▶ Indexes: PK, UQ/FK(ride_id), idx_ride

users

user_id	BIGSERIAL	PK
first_name	VARCHAR(50)	
middle_name	VARCHAR(50)	
last_name	VARCHAR(50)	
email	VARCHAR(100)	
phone_no	VARCHAR(15)	
password_hash	VARCHAR(255)	
role	VARCHAR(20)	
is_active	BOOLEAN	
created_at	TIMESTAMP	

▶ Indexes: PK, UQ(email, phone), idx_role

drivers

driver_id	BIGINT	PK,FK
license_no	VARCHAR(20)	
avg_rating	DECIMAL(3,2)	
total_rides	INTEGER	
is_verified	BOOLEAN	
is_available	BOOLEAN	

▶ Indexes: PK/FK, UQ(license), idx_avail

rides

ride_id	BIGSERIAL	PK
rider_id	BIGINT	FK
driver_id	BIGINT	FK
vehicle_id	BIGINT	FK
fare_id	BIGINT	FK
pickup_address	VARCHAR(200)	
pickup_lat	DECIMAL(10,6)	
pickup_lng	DECIMAL(10,6)	
dropoff_address	VARCHAR(200)	
dropoff_lat	DECIMAL(10,6)	
dropoff_lng	DECIMAL(10,6)	
distance_km	DECIMAL(10,2)	
status	VARCHAR(20)	
estimated_fare	DECIMAL(10,2)	
final_fare	DECIMAL(10,2)	
driver_earning	DECIMAL(10,2)	
platform_commission	DECIMAL(10,2)	
surge_multiplier	DECIMAL(3,2)	
start_pin	VARCHAR(4)	
requested_at	TIMESTAMP	
accepted_at	TIMESTAMP	
completed_at	TIMESTAMP	

▶ Indexes: PK, idx_rider, idx_driver, idx_pending(PARTIAL)

fare_rules

fare_id	BIGSERIAL	PK
category_id	BIGINT	FK
base_fare	DECIMAL(10,2)	
per_km_rate	DECIMAL(10,2)	
per_min_rate	DECIMAL(10,2)	
min_fare	DECIMAL(10,2)	
effective_from	TIMESTAMP	

▶ Indexes: PK, UQ/FK(category_id)

vehicles

vehicle_id	BIGSERIAL	PK
driver_id	BIGINT	FK
category_id	BIGINT	FK
make_model	VARCHAR(100)	
manufacture_year	INTEGER	
plate_no	VARCHAR(20)	
is_active	BOOLEAN	
is_verified	BOOLEAN	

▶ Indexes: PK, UQ(plate), idx_driver, idx_cat

vehicle_categories

category_id	BIGSERIAL	PK
category_name	VARCHAR(20)	
capacity	INTEGER	
description	VARCHAR(200)	

▶ Indexes: PK, UQ(category_name)

riders

rider_id	BIGINT	PK,FK
is_verified	BOOLEAN	
emergency_contact	VARCHAR(15)	

▶ Indexes: PK/FK -> users(user_id)

wallets

wallet_id	BIGSERIAL	PK
driver_id	BIGINT	FK
balance	DECIMAL(12,2)	
currency	VARCHAR(5)	
updated_at	TIMESTAMP	

▶ Indexes: PK, UQ/FK(driver_id)

wallet_transactions

txn_id	BIGSERIAL	PK
wallet_id	BIGINT	FK
ride_id	BIGINT	FK
movement_type	VARCHAR(10)	
purpose	VARCHAR(100)	
amount	DECIMAL(12,2)	
balance_after	DECIMAL(12,2)	
created_at	TIMESTAMP	

▶ Indexes: PK, idx_wallet, idx_ride

vehicle_documents

doc_id	BIGSERIAL	PK
vehicle_id	BIGINT	FK
insurance_policy_no	VARCHAR(50)	
insurance_expiry_date	DATE	
rc_number	VARCHAR(50)	

▶ Indexes: PK, UQ/FK(vehicle_id)

surge_zones

surge_id	BIGSERIAL	PK
zone_name	VARCHAR(100)	
center_lat	DECIMAL(10,6)	
center_lng	DECIMAL(10,6)	
radius_km	DECIMAL(5,2)	
multiplier	DECIMAL(3,2)	
active_from	TIMESTAMP	
active_until	TIMESTAMP	

▶ Indexes: PK, UQ(zone_name)